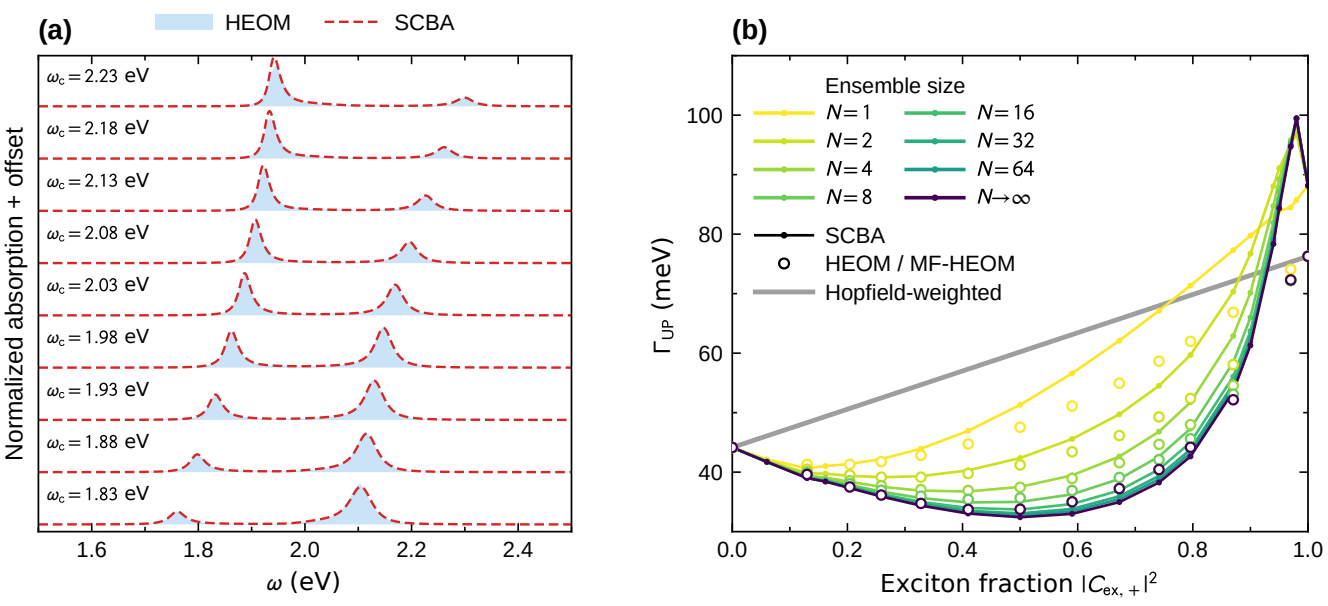

(a)
HEOM
SCBA
$\omega_c = 2.23$ eV
$\omega_c = 2.18$ eV
$\omega_c = 2.13$ eV
$\omega_c = 2.08$ eV
$\omega_c = 2.03$ eV
$\omega_c = 1.98$ eV
$\omega_c = 1.93$ eV
$\omega_c = 1.88$ eV
$\omega_c = 1.83$ eV
Normalized absorption + offset
1.6
1.8
2.0
2.2
2.4
ω (eV)
(b)
Ensemble size
$N = 1$
$N = 2$
$N = 4$
$N = 8$
$N = 16$
$N = 32$
$N = 64$
$N \to \infty$
SCBA
HEOM / MF-HEOM
Hopfield-weighted
Γ_{UP} (meV)
100
80
60
40
0.0
0.2
0.4
0.6
0.8
1.0
Exciton fraction $|C_{ex,+}|^2$